MTH643

Introduction to MATLAB

Virtual University of Pakistan, Lahore

Practice Question for Final Term Lecture (8-15)

Solved by Mahar Afaq Safdar Muhammadi
maharafaq789@gmail.com
03494873115

Lecture # 8

Question: Write the following matrices in MATLAB

$$A=\begin{bmatrix}3 & 4 & 2\\ 2 & 1 & 3\\ 7 & 3 & 8\end{bmatrix} \;\&\; B=\begin{bmatrix}1 & 3 & 9\\ 4 & 2 & 7\\ 3 & 3 & 9\end{bmatrix}$$

- For the matrix equation AX=B, find the matrix X.
- Find the transpose of matrix A.
- Find the inverse of matrix B.
- Find the sum of third row of matrix A and sum of second column of matrix B using direct MATLAB command.
- Find the product of matrix A and B.
- Divide the matrix A by matrix B.

Solution:

```
%Write the following matrices in MATLAB
A=[3,4,2;2,1,3;7,3,8]
B=[1,3,9;4,2,7;3,3,9]

%For the matrix equation AX=B, find the matrix X.
X=inv(A)*B

%Find the transpose of matrix A.
TransposeofA=transpose(A)

%Find the inverse of matrix B.
InverseofB=inv(B)

%Find the sum of third row of matrix A and sum of second column of matrix B
%using direct MATLAB command.
ThirdrowofA=sum(A(3,:))
SecondcolumnofB=sum(B(:,2))

%Find the product of matrix A and B.
ProductofAandB=A*B

%Divide the matrix A by matrix B.
DivideAbyB=A./B
```

Question: Write the following matrices in MATLAB

$$A=\begin{bmatrix}\frac{1}{7} & \frac{2}{3} & 1\\ 4 & \frac{8}{7} & 3\\ \frac{1}{3} & \frac{1}{2} & \frac{1}{8}\end{bmatrix} \; \& \; B=\begin{bmatrix}\frac{1}{6} & \frac{3}{7} & \frac{9}{2}\\ \frac{4}{5} & \frac{2}{3} & \frac{7}{4}\\ \frac{3}{2} & \frac{3}{8} & \frac{9}{4}\end{bmatrix}$$

- For the matrix equation AX=B, find the matrix X.
- Find the transpose of matrix A.
- Find the inverse of matrix B.
- Find the sum of first row of matrix A and sum of third column of matrix B using direct MATLAB command.
- Find the product of matrix A and B.
- Divide the matrix A by matrix B.

Solution:

```
%Write the following matrices in MATLAB
A=[1/7,2/3,1;4,8/7,3;1/3,1/2,1/8]
B=[1/6,3/7,9/2;4/5,2/3,7/4;3/2,3/8,9/4]

%For the matrix equation AX=B, find the matrix X.
X=inv(A)*B

%Find the transpose of matrix A.
TransposeofA=transpose(A)

%Find the inverse of matrix B.
InverseofB=inv(B)

%Find the sum of first row of matrix A and sum of third column of matrix B
%using direct MATLAB command.
FirstrowofA=sum(A(1,:))
ThirdcolumnofB=sum(B(:,3))

%Find the product of matrix A and B.
ProductofAandB=A*B

%Divide the matrix A by matrix B.
DivideAbyB=A./B
```

Lecture # 9

Question

Compute the Eigen values and Eigen vectors for the following matrix using the MATLAB commands

$$\begin{bmatrix} 3 & 1 & 4 \\ 7 & 8 & 9 \\ 3 & 1 & 3 \end{bmatrix}$$

Solution:

```
%Compute the Eigen values and Eigen vectors for the following matrix using
%the MATLAB commands
%Let
A=[3,1,4;7,8,9;3,1,3]
Eigenvalues=eig(A)
[V,D]=eig(A)
```

Question

$$\begin{bmatrix} 3 & 1 & 4 \\ 7 & 8 & 9 \\ 3 & 1 & 3 \end{bmatrix}$$

Find the reduced echelon form of the following matrix using the MATLAB.

Note: Copy the code and also output from MATLAB to exams application.

Solution:

```
%Find the reduced echelon form of the following matrix using the MATLAB.
%Let
A=[3,1,4;7,8,9;3,1,3]
B=rref(A)
```

Output:

```
A =

     3     1     4
     7     8     9
     3     1     3


B =

     1     0     0
     0     1     0
     0     0     1
```

Lecture # 10

Question: Write a *for loop* to add first 20 numbers which are divisible by 5.

Solution:

```
%Write a for loop to add first 20 numbers which are divisible by 5.
v=1:100;
sum=0;
for i=5:5:length(v)
    sum=sum+v(i);
end
disp(sum)
```

Question: Write a program which takes a number as an input from the user and then using the *for loop* display all of its divisors.

Solution:

```
%Write a program which takes a number as an input from the user and then
%using the for loop display all of its divisors.
x=input('Enter a number:');
n=x;
for k=1:n
    if mod(x,k)==0
        disp(k)
    end
end
```

Question: Write a program which compute the factorial of any number.

Solution:

```
%Calculate the factorial of any number.
number=input('Enter a number:');
fact=1;
if number<0
    fprintf('The number you have entered is negative');
else
    for i=1:number
        fact=fact*i;
    end
    disp(fact);
end
```

Question: Write a program in MATLAB that compute the following sum.

$$\sum_{i=1}^{10} 3x^i$$

Solution:

```
%Write a program in MATLAB that compute the following sum.
%[summation i=1:10 (3x^i)].
syms x
sum=0;
for i=1:10
    sum=sum+3*x^i;
end
disp(sum)
```

Question: Write a program in MATLAB that compute the following sum.

$$\sum_{j=1}^{5}\sum_{i=1}^{6} 3^{(i+j)}$$

Solution:

```
%Write a program in MATLAB that compute the following sum.
%[summation j=1:6 summation i=1:5 (3^(i+j)].
sum=0;
for j=1:6
    for i=1:5
        sum=sum+3^(i+j);
    end
end
disp(sum)
```

Question: Write a program which prints all the numbers divisible by 3 and 5 for a given number.

Solution:

```
%Write a program which prints all the numbers divisible by 3 and 5 for a
given number.
x=input('Enter a number: ');
n=x;
for (k=1:n)
    if mod(k,3)==0 && mod(k,5)==0
        disp(k)
    end
end
```

Lecture # 11

Question: Write a function with name 'myfunction' which takes any two numbers as an input and returns the greater number.

Solution:

Editor - E:\bin\myfunction.m

myfunction.m

```
%Write a function with name 'myfunction' which takes any two numbers as an
%input and returns the greater number.
function GreaterNumber=myfunction(x,y)
if x>y
    GreaterNumber=x;
else
    GreaterNumber=y;
end
end
```

Command Window

```
>> x=13;y=17;
>> GreaterNumber=myfunction(x,y)

GreaterNumber =

    17
```

Question: Write a function with name 'newfunction' which takes two numbers as an input and returns the highest common factor of the numbers.

Solution:

Editor - E:\bin\newfunction.m*

newfunction.m*

```
%Write a function with name 'newfunction' which takes two numbers as an
%input and returns the highest common factor of the numbers.
function HCF=newfunction(x,y)
for i=1:1:(x+y)
    if mod(x,i)==0 && mod(y,i)==0
        HCF=i;
    end
end
end
```

Command Window

```
>> x=35;y=75;
>> HCF=newfunction(x,y)

HCF =

     5
```

Question: Write a function with name ‘practicefunction’ which takes any three random numbers as an input and returns the numbers in ascending order.

Solution:

Editor - E:\bin\practicefunction.m*

practicefunction.m*

```
%Write a function with name 'practicefunction' which takes any three random
%numbers as an input and returns the numbers in ascending order.
function AscendingOrder=practicefunction(x,y,z)
A=[x y z];
AscendingOrder=sort(A);
end
```

Command Window

```
>> x=10; y=15; z=12;
>> AscendingOrder=practicefunction(x,y,z)

AscendingOrder =

    10    12    15
```

Question: Write a function with name ‘practicefunction2’ which takes one number as an input and if the number is odd then print ‘This is an odd number’, otherwise print ‘This is an even number’.

Solution:

Editor - E:\bin\practicefunction2.m

practicefunction2.m

```
%Write a function with name ‘practicefunction2’ which takes one number as an
%input and if the number is odd then print ‘This is an odd number’,
%otherwise print ‘This is an even number’.
function EvenOrOdd=practicefunction2(x)
if mod(x,2)==1;
    fprintf('This is an odd number. ');
else
    fprintf('This is an even number. ');
end
end
```

Command Window

```
>> x=13;
>> practicefunction2(x)
This is an odd number. >>
>> x=18;
>> practicefunction2(x)
This is an even number. >>
>> 
```

Lecture # 12

Question: Solve the differential equation $dy/dt=2y+t$ in MATLAB.

Solution:

```
% Solve the differential equation dy/dt=2y+t in MATLAB.
syms y(t)
ode=diff(y,t)==2*y+t;
soly(t)=dsolve(ode)
```

Question: Solve the differential equation $dy/dt=2y+t$ with the initial condition $y(0)=1$.

Solution:

```
% Solve the differential equation dy/dt=2y+t with the initial condition
y(0)=1.
syms y(t)
ode=diff(y,t)==2*y+t;
cond=y(0)==1;
soly(t)=dsolve(ode,cond)
```

Question: Solve the differential equation $dy/dx=2y+x^2$ with the initial condition $y(0)=1$.

Solution:

```
% Solve the differential equation dy/dx=2y+x^2 with the initial condition
y(0)=1.
syms y(x)
ode=diff(y,x)==2*y+x^2;
cond=y(0)==1;
soly(x)=dsolve(ode,cond)
```

Question: Solve the differential equation $d^2y/dx^2+2dy/dx=e^x$ with the initial conditions $y(0)=1, y'(0)=0$.

Solution:

```
% Solve the differential equation d^2y/dx^2+2dy/dx=e^x with the initial
conditions y(0)=1, y'(0)=0.
syms y(x)
Dy=diff(y);
ode=diff(y,x,2)+2*diff(y,x)==exp(x);
cond1=y(0)==1;
cond2=Dy(0)==0;
cond=[cond1 cond2];
soly(x)=dsolve(ode,cond)
```

Lecture # 13

Question: Find the approximate solutions of the following Initial Value Problem on the interval [-3, 3] using ode45 command.

$$y'(x)=y-x^2+5,\ \ y(0)=1$$

Solution:

```
% Find the approximate solutions of the following Initial Value Problem on
the
% interval [-3,3] using ode45 command, y'(x)=y-x^2+5, y(0)=1.
yprime=@(x,y)y-x^2+5;
tspan=[-3 3];
y0=1;
[x,y]=ode45(yprime,tspan,y0)
plot(x,y)
```

Question: Numerically approximate the solution of the following ordinary differential equation using ode45

$$dx/dt = 3e^{-t}$$

Use a time interval of [0, 5] and the initial condition $x(0)=0$.

Solution:

```
% Numerically approximate the solution of the following ordinary differential
% equation using ode45, dx/dt=3e^(-t).
% Use a time interval of [0,5] and the initial condition x(0)=0
xprime=@(t,x)3*exp(-t);
tspan=[0 5];
x0=0;
[t,x]=ode45(xprime,tspan,x0)
plot(t,x)
```

Lecture # 14

Question:
Write the MATLAB program for approximating the following integral using Simpson's 1/3 rule with n =4

$$\int_1^2 \frac{1}{x}dx$$

Solution:

```
% Write the MATLAB program for approximating the following integral using
% Simpson's 1/3 rule with n =4
f=@(x)1/x;
a=1;
b=2;
n=4;
h=(b-a)/n;
sumeven=0;
sumodd=0;
```

```
for k=1:n-1
    x(k)=a+k*h;
    if rem(k,2)==0
        sumeven=sumeven+f(x(k));
    else
        sumodd=sumodd+f(x(k));
    end
end
Area=h/3*(f(a)+f(b)+2*sumeven+4*sumodd)
```

Question:

Write the MATLAB program for approximating the following integral using Simpson's 1/3 rule with n =8

$$\int_1^4 x\sin(x)\,dx$$

Solution:

```
% Write the MATLAB program for approximating the following integral using
% Simpson's 1/3 rule with n =8
f=@(x)x*sin(x);
a=1;
b=4;
n=8;
h=(b-a)/n;
sumeven=0;
sumodd=0;
for k=1:n-1
    x(k)=a+k*h;
    if rem(k,2)==0
        sumeven=sumeven+f(x(k));
    else
        sumodd=sumodd+f(x(k));
    end
end
Area=h/3*(f(a)+f(b)+2*sumeven+4*sumodd)
```

Lecture # 15

Question: Compute the Riemann sum using 12 equally spaced sub-intervals for the following definite integral:

$$\int_0^4 \left(\frac{1}{4}x+2\right)$$

Solution:

```
f=@(x)x/4+2;
a=0;
b=4;
n=12;
h=(b-a)/n;
value=0;
for k=1:n
    c=a+k*h;
    value=value+f(c);
end
Answer=value*h
```

Question: Compute the Riemann sum using 9 equally spaced sub-intervals for the following definite integral:

$$\int_0^3 \frac{x^2}{2}dx$$

Solution:

```
f=@(x)x^2/2;
a=0;
b=3;
n=9;
h=(b-a)/n;
value=0;
for k=1:n
    c=a+k*h;
    value=value+f(c);
end
Answer=value*h
```

Question: Compute the Riemann sum using 5 equally spaced sub-intervals for the following definite integral:

$$\int_0^3 (2x^2+4)dx$$

Solution:

```
f=@(x)2*x^2+4;
a=0;
b=3;
n=5;
h=(b-a)/n;
value=0;
for k=1:n
    c=a+k*h;
    value=value+f(c);
end
Answer=value*h
```

Question: Compute $\int_0^4 x^2 dx$ using Trapezoidal Rule considering 20 equally spaced sub-intervals.

Solution:

```
f=@(x)x^2;
a=0;
b=4;
n=20;
h=(b-a)/n;
sum=0;
for k=1:n-1
    x(k)=a+k*h;
    y(k)=f(x(k));
    sum=sum+y(k);
end
Area=h/2*(f(a)+f(b)+2*sum)
```

Question: Compute $\int_{-1}^3 2^x dx$ using Trapezoidal Rule considering 12 equally spaced sub-intervals.

Solution:

```
syms y
f1=y^2;
Area1=int(f1,a,b);

f=@(x)2^x;
a=-1;
b=3;
n=12;
h=(b-a)/n;
sum=0;
for k=1:n-1
    x(k)=a+k*h;
    y(k)=f(x(k));
    sum=sum+y(k);
end
Area=h/2*(f(a)+f(b)+2*sum)
```

Question: Compute $\int_1^5 \frac{1}{x} dx$ using Trapezoidal Rule considering 16 equally spaced sub-intervals.

Solution:

```
f=@(x)1/x;
a=1;
b=5;
n=16;
```

```
h=(b-a)/n;
sum=0;
for k=1:n-1
    x(k)=a+k*h;
    y(k)=f(x(k));
    sum=sum+y(k);
end
Area=h/2*(f(a)+f(b)+2*sum)
```

Question: Compute the integral $\int_0^5 x(x-2)(x-3)$ using Simpson's Three-Eight Rule by dividing the interval [0, 5] into 10 sub-intervals of uniform length.

Solution:

```
f=@(x)x*(x-2)*(x-3);
a=0;
b=5;
n=10;
h=(b-a)/n;
sum1=0;
sum2=0;
for k=1:n-1
    x(k)=a+k*h;
    if rem(k,3)==0
        sum1=sum1+f(x(k));
    else
        sum2=sum2+f(x(k));
    end
end
Area=3*h/8*(f(a)+f(b)+2*sum1+3*sum2)
```

Question: Compute the integral $\int_{-1}^{1} \tan(x)-2x$ using Simpson's Three-Eight Rule by dividing the interval [-1, 1] into 50 sub-intervals of uniform length.

Solution:

```
f=@(x)tan(x)-2*x;
a=-1;
b=1;
n=50;
h=(b-a)/n;
sum1=0;
sum2=0;
for k=1:n-1
    x(k)=a+k*h;
    if rem(k,3)==0
        sum1=sum1+f(x(k));
```

```
    else
        sum2=sum2+f(x(k));
    end
end
Area=3*h/8*(f(a)+f(b)+2*sum1+3*sum2)
```